

Thursday, December 08, 2011, 03:44 PM

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*
*      STAAD.Pro
*      Version 2006 Bld 1002.US
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date= DEC 8, 2011
*      Time= 15:44:23
*
*      USER ID: RNS Associates
*****
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1. STAAD SPACE
INPUT FILE: 20111208 JRC HANDRAIL.STD
2. START JOB INFORMATION
3. JOB NAME JRC PIG STATION HANDRAIL
4. ENGINEER DATE 08-DEC-11
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 0 0.55 0; 2 0 1 0; 3 0.3 0 0; 4 0.3 0.55 0; 5 0.3 1 0; 6 1.4 0 0
10. 7 1.4 0.55 0; 8 1.4 1 0; 9 2.5 0 0; 10 2.5 0.55 0; 11 2.5 1 0; 12 3.6 0 0
11. 13 3.6 0.55 0; 14 3.6 1 0; 15 3.8 0.55 0; 16 3.8 1 0
12. MEMBER INCIDENCES
13. 100 1 2; 101 1 4; 102 2 5; 103 4 3; 104 5 4; 105 4 7; 106 5 8; 107 7 6
14. 108 8 7; 109 7 10; 110 8 11; 111 10 9; 112 11 10; 113 10 13; 114 11 14
15. 115 13 12; 116 14 13; 117 13 15; 118 14 16; 119 15 16
16. DEFINE MATERIAL START
17. ISOTROPIC STEEL
18. E 2.05E+008
19. POISSON 0.3
20. DENSITY 76.8195
21. ALPHA 1.2E-005
22. DAMP 0.03
23. END DEFINE MATERIAL
24. MEMBER PROPERTY BRITISH
25. 100 TO 102 105 106 109 110 113 114 117 TO 119 TABLE ST PIP484.0
26. 103 104 107 108 111 112 115 116 TABLE ST PIP484.0
27. CONSTANTS
28. MATERIAL STEEL ALL
29. SUPPORTS
30. 3 6 9 12 FIXED
31. LOAD 1 LOADTYPE NONE TITLE DL
32. SELFWEIGHT Y -1
33. LOAD 2 LOADTYPE NONE TITLE LL
34. MEMBER LOAD
35. 102 106 110 114 118 UNI GZ 0.814
36. 102 106 110 114 118 UNI GY -0.6
37. LOAD COMB 3 1.0DL + 1.0LL
38. 1 1.0 2 1.0
39. LOAD COMB 4 1.4DL + 1.6LL
40. 1 1.4 2 1.6
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41. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

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NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 16/ 20/ 4
ORIGINAL/FINAL BAND-WIDTH= 3/ 3/ 18 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 72
SIZE OF STIFFNESS MATRIX = 2 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 100700.4 MB
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42. PRINT SUPPORT REACTION ALL

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
3	1	0.00	0.14	0.00	0.00	0.00	0.00
	2	0.00	0.50	-0.70	-0.74	0.02	0.00
	3	0.00	0.64	-0.70	-0.74	0.02	0.00
	4	0.00	0.99	-1.12	-1.19	0.03	0.00
6	1	0.00	0.14	0.00	0.00	0.00	0.00
	2	0.00	0.67	-0.89	-0.84	0.00	0.00
	3	0.00	0.81	-0.89	-0.84	0.00	0.00
	4	0.01	1.26	-1.42	-1.34	0.01	0.00
9	1	0.00	0.14	0.00	0.00	0.00	0.00
	2	0.00	0.67	-0.89	-0.83	-0.01	0.00
	3	0.00	0.81	-0.89	-0.83	-0.01	0.00
	4	0.00	1.27	-1.42	-1.32	-0.02	0.00
12	1	0.00	0.12	0.00	0.00	0.00	0.00
	2	0.00	0.43	-0.62	-0.69	-0.03	0.00
	3	0.00	0.56	-0.62	-0.69	-0.03	0.00
	4	0.00	0.87	-0.99	-1.10	-0.05	0.00

***** END OF LATEST ANALYSIS RESULT *****

43. LOAD LIST 4

44. PRINT MEMBER FORCES ALL

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MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN			METE	(LOCAL)				
MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
100	4	1	-0.01	0.01	-0.06	0.02	0.09	0.00
		2	0.04	-0.01	0.06	-0.02	-0.07	0.00
101	4	1	0.01	0.01	0.06	0.09	-0.02	0.00
		4	-0.01	0.00	-0.06	-0.09	0.01	0.00
102	4	2	-0.01	-0.04	-0.06	-0.07	0.02	0.00
		5	0.01	0.35	-0.33	0.07	-0.06	-0.05
103	4	4	0.96	0.00	1.12	0.03	0.57	0.00
		3	-0.99	0.00	-1.12	-0.03	-1.19	0.00
104	4	5	0.89	-0.11	1.07	0.05	-0.03	-0.04
		4	-0.92	0.11	-1.07	-0.05	-0.46	-0.01
105	4	4	-0.10	0.04	0.01	-0.02	-0.02	0.01
		7	0.10	0.03	-0.01	0.02	0.01	0.00
106	4	5	0.10	0.55	-0.74	-0.04	0.11	0.09
		8	-0.10	0.58	-0.69	0.04	-0.08	-0.11
107	4	7	1.23	-0.01	1.42	0.01	0.55	0.00
		6	-1.26	0.01	-1.42	-0.01	-1.34	0.00
108	4	8	1.14	0.00	1.41	0.01	-0.05	0.00
		7	-1.17	0.00	-1.41	-0.01	-0.58	0.00
109	4	7	-0.10	0.04	0.00	0.01	-0.01	0.01
		10	0.10	0.03	0.00	-0.01	0.02	0.00
110	4	8	0.10	0.56	-0.72	0.01	0.10	0.11
		11	-0.10	0.56	-0.72	-0.01	-0.10	-0.10
111	4	10	1.24	0.00	1.42	-0.02	0.54	0.00
		9	-1.27	0.00	-1.42	0.02	-1.32	0.00
112	4	11	1.14	-0.03	1.39	-0.03	-0.05	-0.01
		10	-1.17	0.03	-1.39	0.03	-0.57	-0.01
113	4	10	-0.13	0.04	-0.03	0.04	-0.01	0.01
		13	0.13	0.03	0.03	-0.04	0.04	0.00
114	4	11	0.13	0.58	-0.68	0.06	0.07	0.11
		14	-0.13	0.54	-0.76	-0.06	-0.11	-0.08